

2

Theoretical Concepts

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Mesoscopic systems exhibit quantum transport phenomena. There, the sample's size becomes comparable or lower than the characteristic lengths for electron travelling, and, as a result, physical properties such as the electrical conductivity depend on the specimen's dimensionality.

Typically, three characteristic lengths are investigated in mesoscopic transport experiments. The mean free path ℓ , the wavelength of Fermi λ_F , and the phase-relaxation length L_φ . The mean free path is the average distance an electron travels prior to being dispersed by a scattering centre; attempts to measure it

Table 2.1: Physical parameters relevant in transport phenomena. See also [44, 140]

<i>Material type</i>	n (cm ⁻³)/[cm ⁻²]	λ_F (nm)	ℓ (nm)	L_φ (μm)
Metal	10 ²³	0.5	1 – 10	0.5
Graphene	[10 ¹¹]	$2\sqrt{\pi/n}$	50 – 3,000	0.5
SW-CNT	[10 ¹¹]	0.74	1,000	3
MW-CNT	–	–	10 – 40	0.1

for graphene nanoribbons are briefly described in Ch 1. The Fermi wavelength $\lambda_F = 2\pi/k_F$ is the wavelength of an electron at Fermi energy E_F , whereas the phase-relaxation length is the average distance over which an electron travels before its quantum-mechanical phase becomes randomised.

2.1 Why an Exchange of Energy Matters

If the scattering of an electron is *elastic*, such as when the scattering event does not induce any electronic transition between the ground state and any excited states, the wavefunction undergoes essentially a phase shift following the event, remaining coherent. Therefore, an elastic event may only affect ℓ , leaving L_φ unchanged. In contrast, if the electron wave loses energy by either excitation of core electrons, spins associated with magnetic impurities, or phonons, both the mean free path and the phase-relaxation length may be affected by such (*inelastic*) scattering events. Tab 2.1 summarises a few of these factors for various materials as a function of carrier concentration n .

Because only electrons around the Fermi energy E_F affects the transport properties, the mean free path and phase-relaxation length are often scaled by the Fermi velocity

$$t_m = \frac{L_m}{v_F}, \quad \text{and} \quad t_\varphi = \frac{L_\varphi}{v_F},$$

The introduced t_m and t_φ are, respectively, the momentum relaxation time and phase-relaxation time, and are often compared to the time it takes one electron to travel the length of the sample. However, these two numbers do not correlate to

the period of a single scattering event by the relevant scattering mechanism; rather, they represent the average time for the several collisions that result in significant changes of momentum and phase, respectively. In this context, significant entails, for instance, that the sums of the changes in momentum and phase over time t_m and t_φ reach $\hbar k_F$ and π , respectively [141].

There are several ways through which it is possible to control the various scattering mechanisms. For example it is possible to suppress the magnetic scattering by applying a magnetic field [142, 143], whilst phonons excitations might be suppressed at low temperature, although it is the electron-phonon interaction which controls the relaxation of an electron by an emitting a phonon [144]. The electron-electron interaction is determined by the carrier concentration and the extension of the valence band along the Brilluoin zone [145]. On top of that, all the frequencies corresponding to the various scattering processes might be changed by controlling the Fermi velocity, often obtained by either varying a gate voltage or by changing the conduction band curvature.

2.2 Electronic Transport Regimes

In charge-transport experiments, molecules are typically a few nanometer in size, orders of magnitude smaller than what tweezers or other common manipulation tools can handle. Since the number of lateral confinements imposed on these system is three dimensional, the corresponding DOS approaches the Dirac delta distribution $\rho_{0D} \sim \delta(E)$.

An analogous platform where the density of state approaches the Dirac delta functions are quantum dots (QDs), historically fabricated on a metallic substrate through lithography means (Fig. 2.1), and capable of hosting electrons. For these platforms, a wide breadth of theoretical and experimental evidences have been accumulated since the 1980, with application ranging from the nanoelectronics to quantum computing. While source and drain are electrically contactd to the central island, one or more gates are just capacitatively coupled.

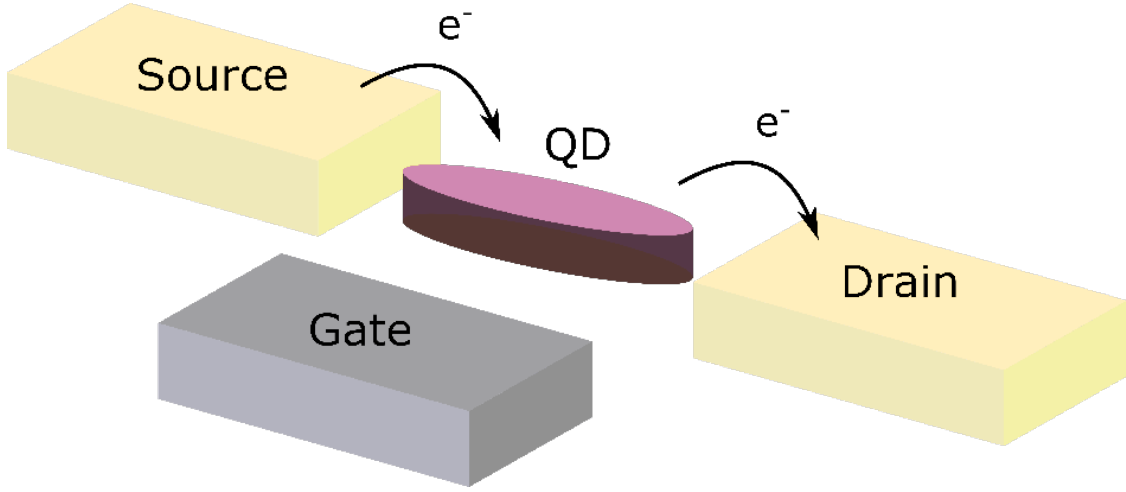


Figure 2.1: Cartoon schematic for a single-electron transistor based on a quantum dot.

One way to describe this system is through the constant interaction model, which allows to derive important electrical transport information within the *orthodox* Coulomb framework. Within this model the Coulomb interactions both between electrons on the dot and between the dot and the leads, are described by a single capacitance C which stays constant and it is given by the total sum of the individual capacitances between the dot with its terminals

$$C = \sum_{i=1}^{\text{terminals}} C_i, \quad (2.1)$$

where i runs over all the terminals coupled to the central island. Secondly, the total Hamiltonian may be modelled as a sum of single-particle Hamiltonians, meaning that the total energy of the system can be expressed as a sum of single-particle energies $\Delta\epsilon$ which corresponds to the individual, non-interacting energy levels. Under these assumptions, the total dot energy given for a quantum dot in a three-terminal device with N electrons becomes

$$U(N) = \sum_i^N E_i + \frac{[e(N - N_0) + C_S V_S + C_D V_D + C_G V_G]^2}{2C}, \quad (2.2)$$

where eN_0 is the charge present on the dot when $V_{SD} = V_G = 0$ due to positive background charges in the substrate. One of the central quantity to evaluate whether

the current can flow through the dot is the chemical potential, which is the minimum energy required to add an extra electron to the dot, and may be expressed as

$$\begin{aligned}\mu_{\text{dot}}(N) &= E_{\text{dot}}(N) - E_{\text{dot}}(N-1), \\ &= \left(N - N_0 - \frac{1}{2}\right) E_C - \frac{E_C}{e} (C_S V_S + C_D V_D + C_G V_G) + E_N,\end{aligned}\quad (2.3)$$

where $E_C = e^2/C$ is the charging energy. In the simplest case as the one describe, it is the interplay between this energy, the single-particle level spacing $\Delta\epsilon$, the coupling Γ between the leads and the dot, and the thermal energy $k_B T$ that determine the overall transport regime. In what follows, I will briefly outlined the different transport regimes at the nanoscale based on the relative magnitude of these energy scales.

2.2.1 Weak Coupling Regime

The criteria that has to be satisfied in order to operate in a weak coupling regime is that the lifetime broadening of a molecular state has to be the smallest energy scale in place, i.e. $\Gamma \ll E_C, E_{\text{add}}(N), k_B T$. This condition is often realised in an electrical break junctions such as the ones used in this thesis.

In this regime, Coulomb blockade (Fig. 2.2) and single-electron (sequential) tunnelling prevails. This transport mechanism involves two tunneling steps. Firstly, an electron tunnels from one electrode into the dot (or vice versa), thereby changing the overall charge of the dot, e.g. from $|N\rangle$ to $|N+1\rangle$; during this event the spin also changes by $\pm 1/2$. Subsequently, the electron tunnels from the molecule to the second reservoir. We can write this sequence of SET events as

$$|N\rangle \leftrightarrow |N+1\rangle \leftrightarrow |N\rangle \quad (2.4)$$

This process is only allowed when the electrochemical potential (or Fermi level) of the first electrode E_{F1} is higher than the electrochemical potential of the nano-object μ_N and the Fermi level of the second electrode E_{F2} is lower than μ_N . The reverse process is also possible, hence the double-ended arrow in Eq. 2.4.

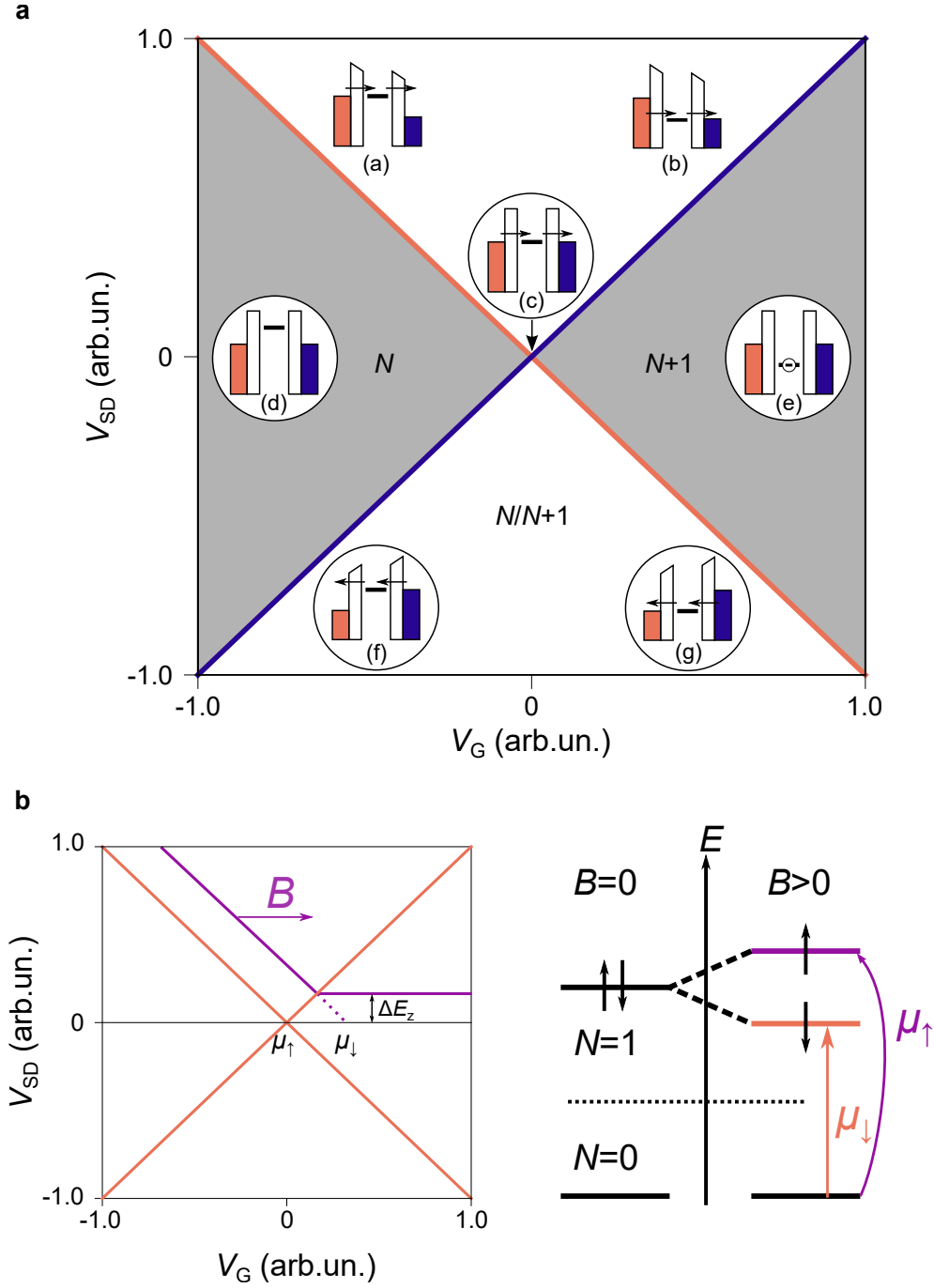


Figure 2.2: **a**, Coulomb blockade schematic. Charge is fixed in grey regions. This is also represented by (d) and (e) where the quantum dot energy level either lies outside the V_{SD} window or is doubly-occupied, respectively. Electron tunnelling through the dot is allowed within the white regions, and all the possible cases are detailed (a–c and f–g). **b**, **left** Ground and excited state separated in energy by the Zeeman interaction. Conductance step may occur when $e|V_{SD}| = \Delta E_{\text{Zeeman}}$ in the blockade region. **b**, **right** Two different charge states. Degeneracy between spin up and spin down is lifted by the external magnetic field B .

By either applying a bias $eV_{\text{SD}} = E_F(1) - E_F(2)$ over the two electrodes or tuning μ_N through a gate electrode is possible to switch on this regime. Electron transport in the SET regime is schematically depicted in Fig. 2.2a, which shows an electrochemical potential scheme with two electron reservoirs filled up to their Fermi levels, $E_F(1)$ and $E_F(2)$. The schematic also shows SET events where one electron tunnels from one of the electrode into the dot depending whether the bias polarity is positive or negative. These kind of way to visualise the current (or the conductance) as a function of gate and bias voltages is called *charge stability diagram*. The Coulomb blockade (grey regions) are characterised by very low current levels, and are often referred to as Coulomb diamonds. In these regions transport is blocked because the energy cost to add an electron onto the QD exceeds the available energy.

Lines running parallel to diamond edges originate from orbital, spin, or vibrational excited states. These contribute to the current flow through the dot whenever their corresponding energy levels are included in the bias window. The investigation of such states often provide important information about the dot energy spectrum, and are particularly useful when attempting to identify specific molecules, as often these have a unique and distinctive vibrational fingerprint.

2.2.2 Strong Limit

When the lifetime broadening of a molecular/dot state is greater than all the other energy scales at play, i.e. $\Gamma \gg E_C, E_{\text{add}}(N), k_B T$, a strong hybridisation between lead and dot state occurs, resulting in a suppression of Coulomb blockade by the quantum fluctuations of the electric charge. In this context the dominant transport mechanism is elastic coherent tunnelling, and the electrical conductance may be approximated as $G \simeq \frac{2e^2}{h} \mathcal{T}(E)$, where we have indicated with $\mathcal{T}(E)$ the transmission probability for all the available conduction channels through the quantum dot, evaluated at the Fermi energy.

2.2.3 Intermediate Limit

Lastly, the intermediate coupling regime has attributes of both limits, it still exhibits a noticeable Coulomb blockade, but also shows transport within the zero current region. The two most noticeable new transport channels available in this regime are elastic and inelastic cotunnelling: these higher-order transport mechanisms involve the simultaneous tunnelling of two electrons. Initially, the QD is in a state with an unoccupied excited state still below the electrochemical potential of the leads, in contrast to resonant tunnelling where the excited state is between the source and drain electrochemical potential. When the electron tunnels out of the QD onto the drain, the QD is left in a virtual state and there are two possibilities for the orbit to be refilled. First, the new electron simultaneously tunnelling in from the source lead occupies the ground state again, therefore leaving the QD in exactly the same configuration as before. This is called elastic cotunnelling. If the new electron, however, tunnels into the excited state, the QD has changed its energy and the process is called inelastic cotunnelling. The electron can subsequently relax into the ground state, rendering the QD again in the initial configuration. Importantly, for inelastic cotunnelling to take place, a bias larger than the excitation energy of the dot has to be applied and since the QD excitation is created at the expense of the bias applied, electron energy is conserved. Overall, both elastic and inelastic cotunnelling lead to conduction in the blocked Coulomb diamond region.

No clear dominance from one of the energy scale is present in this transport regime. As we have outlined in Sec. 2.4, charge transport is largely suppressed outside the single-electron tunnelling regions. There are instances, however, where an electron from one lead might still be able to reach the other lead through a process called cotunnelling (COT).

Outside of the SET regime, i.e., where the requirements in Equation (2.1) are not met, SET events are forbidden and charge transport is largely suppressed (see blue regions in Figure 2.1b). Electrons from one electrode can still reach the other electrode via the nanoobject through a process known as cotunneling (COT). In the simplest form of COT, two electrons are involved in a single COT event. We

consider the case where $\mu_N < E_{F1}, E_{F2}$, such that an electron from the nano-object cannot tunnel to the electrodes by an SET event. We refer to the difference between μ_N and E_{F1} (or E_{F2}) as ε . In the case of cotunneling, an electron can tunnel into the electrodes, as long as another electron tunnels back into the nano-object within a very short time. Energy conservation is briefly violated in this process, which is quantum-mechanically allowed within a timescale $\sim \hbar/\varepsilon$. A COT event is schematically depicted in Figure 2.1c in which (1.) an electron tunnels from the nano-object to electrode 2 and (2.) an electron from electrode 1 tunnels to the nano-object. In the depicted case, the energy of the initial and the final state is the same and this process is therefore called elastic COT.

If the energy of the initial state is different than the energy of the final state, one speaks of inelastic COT. Examples of inelastic COT include transitions to excited spin states and excited vibrational states. These inelastic processes only contribute to transport when the bias voltage exceeds the energy difference between the initial and final state. Accordingly, in a transport measurement a step in the dI/dV can be expected at the bias voltage corresponding to this excitation energy. The position at which the dI/dV step occurs is indicated by the horizontal dashed lines in the COT region of Figure 2.1b. From the COT dI/dV excitation steps one can extract information about the spin- and vibrational excitations in a system. In the case of spin excitations, the only allowed transitions are given by the COT spin-selection rules [1]:

$$\Delta S = 0, \pm 1 \text{ and } \Delta m = 0, \pm 1,$$

in which ΔS and Δm are the differences in the total spin and the magnet quantum numbers, respectively, between the initial and the final state. This requirement for COT follows from the fact that only two spin-1/2 particles - the two electrons - are involved in the COT process.

2.3 Landauer Theory

In order to make predictions about the electronic and thermoelectric properties of QDs, a good model for electronic transport in mesoscopic to molecular-scale devices is necessary. In the limit of predominantly SET and assuming coherent transport, extending the Landauer formalism (originally a model for electrical conduction) towards thermoelectric effects has proven successful and is widely used. The Landauer formalism is a concept formulated in 1947 by Rolf Landauer that models conduction as transmission. The basic assumption is that two electron reservoirs are connected by a scatterer (QD island for the transport measurements presented here) through which electron transport is possible. The transport is then determined by the probability of electrons moving from one reservoir to the other through the scatterer. This probability is always between 0 and 1, which means there is an upper limit on conductance.

Employing the Landauer approach, the current through such a structure can be written as:

$$I = \frac{2e}{h} \int_{-\infty}^{+\infty} \mathcal{T}(\epsilon) [f_L(E) - f_R(E)] d\epsilon, \quad (2.5)$$

where $\mathcal{T}$ is the transmission probability of an electron with single-particle energy ϵ to move across the source and drain leads. The Fermi-Dirac statistics $f_{\text{FD}}(\epsilon, T) = \frac{1}{1+e^{\epsilon_i - \mu/k_B T}}$ details the average number of fermions (electrons in our case) in a single-particle state i with energy ϵ_i in thermodynamic equilibrium at temperature T . Therefore, it is used to quantify how many electrons from the source leads have a certain energy to be able to cross the molecular junction: this event, as everything in quantum mechanics, is detailed by the amplitude of the transmission probability $\mathcal{T}(\epsilon)$.

One can calculate the number of electrons in a crystal by multiplying the density of states $N(\epsilon)$ with the FD distribution and summing up this product for all energies:

$$n = \int_0^{\infty} f_{\text{FD}}(\epsilon) N(\epsilon) d\epsilon, \quad (2.6)$$

The density of states $N(\epsilon)$ denotes the number of electron states per unit of energy and volume and differs for different materials. In such a system the chemical potential is defined as the amount of energy needed to add an additional electron to the system and naturally this quantity has an inherent temperature dependence as the energy of the system changes with T . An approximation for the chemical potential as a function of temperature and energy has been developed using the Sommerfeld expansion (cf. Appendix B): since we are interested in low-temperature properties, we will consider only terms up to the second order of Eq. B.7:

$$\mu \simeq \epsilon_F - \frac{\pi^2}{6} (k_B T)^2 \frac{1}{N(\epsilon_F)} \frac{\partial N(\epsilon)}{\partial \epsilon} \Big|_{\epsilon=\epsilon_F} \quad (2.7)$$

As is evident from this equation, the chemical potential decreases as the temperature rises, resulting in a potential gradient whenever a temperature gradient is applied. Any difference in T will therefore lead to a flow of electrons as the cost of adding another electron to the hot side would be lower than adding it to the cold side. This results in the build-up of a thermoelectric voltage in an open circuit.

By multiplying this probability with the elementary charge e gives the current for one energy level with a factor of two added due to spin degeneracy. After adding all of these possible transport channels together, which is done by taking the integral over all energies, and accounting for dimension normalization via the Planck constant h it is possible to calculate the total current I across the junction.

The electrical conductance is then retrieved by taking the derivative of the current:

$$\begin{aligned} G &= -\frac{2e}{h} \int_{-\infty}^{+\infty} \mathcal{T}(\epsilon) \frac{\partial f(T, \epsilon)}{\partial \epsilon} d\epsilon, \\ &= -G_0 \int_{-\infty}^{+\infty} \mathcal{T}(\epsilon) \frac{\partial f(T, \epsilon)}{\partial \epsilon} d\epsilon, \end{aligned} \quad (2.8)$$

Several computational scheme may be used to compute the set of transmission eigenvalues for a given molecular hamiltonian, ranging from simple tight-binding to non-equilibrium Green's functions. The interested reader is referred to the specific literature [146].

2.4 Accounting the Vibrational Degree of Freedom: the Rate-Equation Model

The rate equation model starts from a kinetic description of the electrode–molecule junction and assumes the overall current flowing through the molecule uniquely determine by how fast an electrons tunnels in and out from a weakly coupled molecular level [147, 148]. A promising extension of this framework attempts to quantify the quantum mechanical effects of the vibrational degree of freedom (DOF) on the electron transfer rates. Here we will adopt the description reported in references [149] and [150, 151] where the electronic degrees of freedom interact with an unstructured environment modelled by a superohmic bath having an exponential cut-off:

$$\mathcal{J}(\omega) = \frac{\lambda \omega^3}{2 \omega_c} e^{-\omega/\omega_c}, \quad (2.9)$$

where λ is the reorganization energy as defined from the Marcus model for electron transfer in solution and ω_c is a cut-off frequency.

The current through the junction can be written as

$$I = |e| \frac{\gamma_{\text{ox}}^{\text{S}} \gamma_{\text{red}}^{\text{D}} - \gamma_{\text{red}}^{\text{S}} \gamma_{\text{ox}}^{\text{D}}}{\gamma_{\text{red}}^{\text{S}} + \gamma_{\text{red}}^{\text{D}} + \gamma_{\text{ox}}^{\text{S}} + \gamma_{\text{ox}}^{\text{D}}}, \quad (2.10)$$

where $\gamma_{\text{red/ox}}^{\text{S/D}}$ are the electron-transfer rates at each electrode corresponding to a reduction or an oxidation. If one consider the charge transition from N to the $N + 1$ charge state, the electron rates are given as follows:

$$\gamma_{\text{red}}^{\text{S,D}} = 2 \frac{\Gamma_{\text{S,D}}}{\hbar} \int f_{\text{S,D}}(\epsilon) k_{\text{red}}(\epsilon) d\epsilon, \quad (2.11)$$

$$\gamma_{\text{ox}}^{\text{S,D}} = \frac{\Gamma_{\text{S,D}}}{\hbar} \int [1 - f_{\text{S,D}}(\epsilon)] k_{\text{ox}}(\epsilon) d\epsilon, \quad (2.12)$$

where $\Gamma_{\text{S,D}}$ is the electronic coupling between molecule and the electrodes having Fermi-Dirac distributions $f_{\text{S,D}}(\epsilon)$. The density of states for the reduced and oxidized might be written as:

$$k_{\text{red/ox}}(\epsilon) = \frac{1}{\pi} \text{Re} \int_0^\infty e^{i(\epsilon - \epsilon_0)t/\hbar} e^{-t/\tau} B(t) dt, \quad (2.13)$$

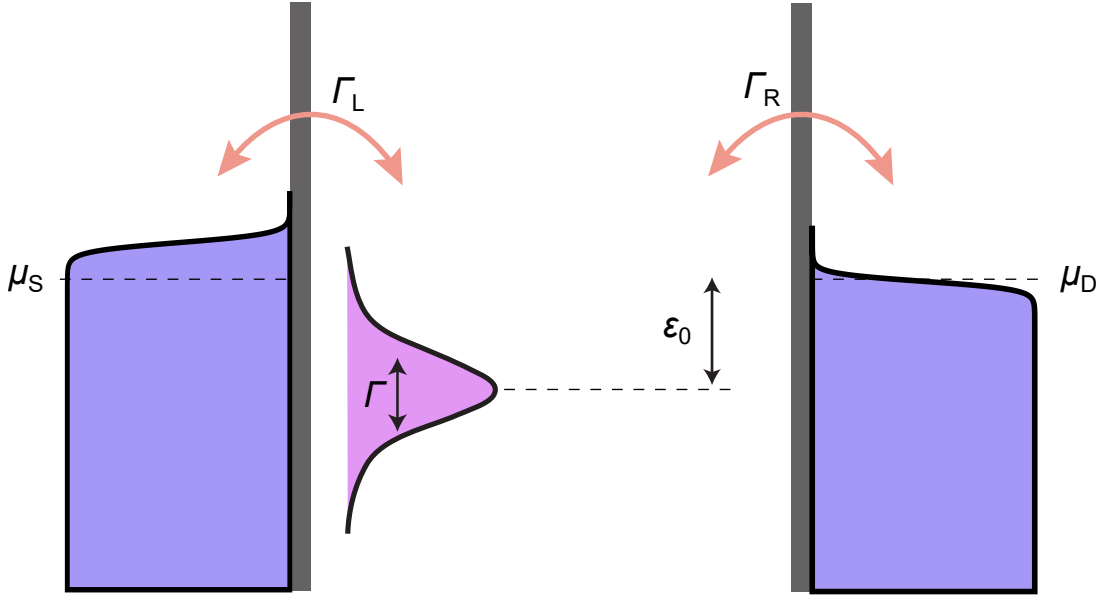


Figure 2.3: Schematic for a single-level transport model. The single-particle states having energy ϵ_0 and lifetime Γ is the closest to the Fermi level of source μ_S and drain μ_D , and solely contributes to the total current flowing across the junction. Γ_L and Γ_R are the tunnelling coupling with the left and right electrode, respectively.

where $\sigma = 1(-1)$ for reduction (oxidation) process, ϵ_0 is the molecular energy level, $\tau = 2\hbar/(\Gamma_S + \Gamma_D)$ is the lifetime of the molecular electronic state. The term written as $B(t)$ is the phononic correlation function t :

$$B(t) = e^{D_q(t)} \cdot e^{D_{\text{bath}}(t)}, \quad (2.14)$$

Here we have written the phononic correlation function $B(t)$ as the product between a single-mode term and a mean-field contribution given by the environment. The two terms might be expressed as follows [152]:

$$D_q(t) = \left(\frac{g_q}{\omega_q} \right) \left[(\cos \omega_q t - 1) \cdot \coth \left(\frac{\omega_q}{2k_B T} \right) - i \sin(\omega_q t) \right], \quad (2.15)$$

$$D_{\text{bath}}(t) = \int \frac{J(\omega)}{\omega^2} \left[(\cos \omega t - 1) \cdot \coth \left(\frac{\omega}{2k_B T} \right) - i \sin(\omega t) \right] d\omega, \quad (2.16)$$

where D_q is the single-mode contribution and D_{bath} is the contribution from the superohmic bath. g_q is the electron-vibration coupling strength, ω is the vibron

frequency and the spectral density, $J(\omega)$, of the phonon bath with reorganisation energy, λ , and cut-off frequency, ω_c , is

We use this model in Chapter 6 to reproduce the low temperature transport features of a molecular graphene nanoribbon with a strong electron–vibron coupling.

2.5 Basics of One-dimensional Physics

In this section we will very briefly review the Fermi liquid theory and its failure in describing strongly-correlated 1D systems. As a result of the lateral confinement of the carriers, graphene nanoribbons are indeed a quasi-1D system. The term *quasi* refers to the electron’s confinement in two directions, leaving only one DOF. Not until the introduction of bottom-up techniques, which are capable of manufacturing atomically-precise ribbon widths below 5 nm, did this become reality.

The topics introduced here will explain the peculiarities in the density of states for a fermionic system and later be used to derive the tunneling density of states in the presence of long-range Coulomb interactions for a one-dimensional chain. These results are central to explain a few experimental results exposed in Chapter 5. A more exhaustive treatment of the following concepts might be reviewed in [153–156].

2.5.1 A Succint Primer of Fermi-Liquid Theory and Its Inadequacies

The Fermi liquid theory was advanced by Lev Landau in 1956 as a phenomenological theory to describe the low-temperature limit behaviour of liquid ^3He heat capacity, which is surprisingly similar the ideal gas one. Since then, the theory has been used and extend also to capture many-body effects in fermionic systems, such as conducting electrons.

The ground state of a non-interacting gas may be fully captured following standard statistical mechanics. In this framework, of a great importance in calculating macroscopic quantities is the gas momentum distribution function n_k^0 , which depends on the wave number $\vec{k}$. More generally, the ground state depends also from the spin occupation number σ but we will drop the spin subscript for

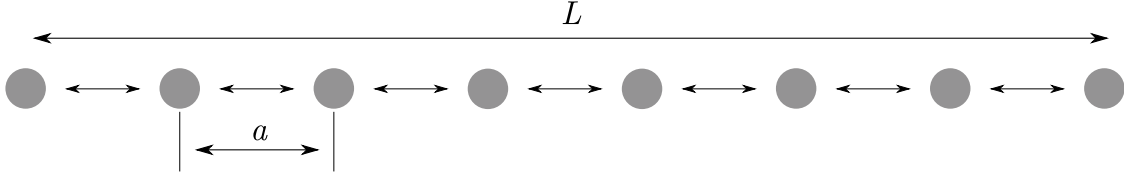


Figure 2.4: Cartoon representing a 1D electron system. The density of the system is defined as the number of particles N in confined in a 1D box of length L , i.e. $\rho = N/L = 1/a$, where a is the inter-particle spacing.

the moment. For a fermionic gas, the zero-temperature limit n_k^0 is the well-known step function which equals to:

$$n_k^0 = \begin{cases} 1 & k \leq k_F \\ 0 & k > k_F \end{cases} \quad (2.17)$$

At finite temperature, the momentum distribution function coincides with the Fermi–Dirac distribution for metals and with the Bose–Einstein distribution for bosons. The energy of a free particle follows the well-known parabolic dispersion, $\epsilon_{\vec{k}} = \hbar k^2/m$, and the total (multi-electron) energy change caused by an excitation might be expressed as a functional of the energy deviation from the ground state energy E_0 :

$$\delta E[\delta n] = E[n] - E_0 = \sum_{\vec{k}} \epsilon_{\vec{k}} \delta n_{\vec{k}} \quad (2.18)$$

Thus the single-particle energy might be written as a functional derivative of the total energy with respect to the total number of particles with a specific $\vec{k}$ vector:

$$\epsilon(\vec{k}) = \frac{\delta E[n]}{\delta n_{\vec{k}}} \quad (2.19)$$

An elementary excitation can therefore be thought as the process of adding/removing a particle to/from the ground state and an arbitrary excited state may consist of a number of such particle-hole pairs. Close to the Fermi surface, $\vec{k} = k_F$, the energy dispersion is sufficiently linear enough to apply a Taylor series expansion:

$$\epsilon_{\vec{k}} = \epsilon_F + v_F \hbar (k - k_F) + \mathcal{O}((k - k_F)^2) \quad (2.20)$$

where $v_F = \nabla_{\vec{k}} \epsilon|_{\vec{k}=k_F}$ is the Fermi velocity. Landau's intuition was assuming that the low-lying eigenstates of the non-interacting Fermi gas continuously evolve into eigenstates of the multi-body (real) system as the interaction is adiabatically switched on. This process leads to the recovery of an individual excitation description, called quasiparticle, which is the starting point of the Fermi-liquid theory. Pictorially, a quasiparticle can be visualised as a fermion surrounded by a cloud of disturbance arises due to interaction with other particles. As a consequence, the mass is replaced by to effective mass $m^* = \hbar k_F / v$ that captures the strenght of the interaction cloud with the single particle.

Therefore, the quasiparticle dispersion may be written as:

$$\epsilon_{\vec{k}} = \epsilon_F + v \hbar (k - k_F) + \mathcal{O}((k - k_F)^2) \quad (2.21)$$

In order to quantify the quasiparticle excitation energies, the energy functional has to be expanded to the second order through a Taylor series expansion as I did before:

$$\delta E[\delta n] = \sum_{\vec{k}} \epsilon_{\vec{k}} \delta n_{\vec{k}} + \frac{1}{2V} \sum_{\vec{k}} f(\vec{k}, \vec{k}') \delta n_{\vec{k}} \delta n_{\vec{k}'} \quad (2.22)$$

where the intensive quantity:

$$\frac{f(\vec{k}, \vec{k}')}{V} = \frac{\delta^2 E[n]}{\delta n_{\vec{k}} \delta n_{\vec{k}'}} \quad (2.23)$$

is known as Landau function and quantifies the interaction strength between quasiparticles. The energy of a quasiparticles can be Similarly defined as

$$\tilde{\epsilon}_{\vec{k}} = \frac{\delta E[\delta n]}{\delta n_{\vec{k}}} = \epsilon_{\vec{k}} + \frac{1}{V} \sum_{\vec{k}'} f(\vec{k}, \vec{k}') \delta n_{\vec{k}'} \quad (2.24)$$

.

So why the Fermi liquid theory fails to capture one-dimensional systems? Essentially because the Landau function is treated by means of a mean field approximation and a mean field kind of description breaks down in 1D [153]. This is caused by several reasons.

Firstly, for a system arranged in a linear chain fashion, only collective excitations exist. Pictorially, one might convince herself by thinking of an electron which starts to oscillate around its equilibrium position. Due to the strong electron–electron interactions the motion is fastly transmitted all along the chain, resulting in a collective excitation. Secondly, in high dimensions we have seen that even if the particles interact with each other one can still recover an individual excitation description, where the interaction results in the concept of quasiparticles.

All these conditions break down for quantum systems in 1D because the one dimensionality introduces first strong quantum fluctuations that destroys the mean phase of your systems (for example in mean field theories one has $\psi = |\psi|e^{i\phi}$ and you replace the phase ϕ with an averaged one).

2.5.2 Introduction to the Tomonaga-Luttinger Model

Several peculiarities encompass one-dimensional systems. As already mentioned, 1D excitations for an interacting fermionic systems are hydrodynamic-like modes—similar to sound waves—which are split into charge- and spin-density waves [155]. The absence of an efficient charge screening leads to long-range interactions, which cause strong correlations and the break down of an independent particle description. These presence of such strong correlations are the main reason why in one-dimensional systems one is required to compute correlation functions for calculating quantity of interest. From here, the wide employment of Green’s functions (propagators).

One consequence of these characteristics is that—very conveniently— in 1D systems it is often sufficient to conduct an asymptotic analysis for characterising your system. In particular, one has to provide the wave velocity u and the interaction parameter K (also called g), which is a dimensionless number which controls the decay of all the correlation functions as well as the thermodynamic quantities [154].

The first model proposed in 1950 by Tomonaga aimed to describe the multi-fermionic ground state of an interacting Fermi gas and its excitations by treating them as solutions to field equations describing sound waves, which are bosonic [157]. The model was later extended by Luttinger in 1963 in an attempt to include

arbitrary interaction strengths and range (short and long) [158]. The correct solution was firstly advanced by Mattis and Lieb in 1965 [159] and finalised by a genial intuition from Haldane in 1981 [160].

A central tool required to arrive to Haldane's solution is bosonisation, a mathematical technique which maps the excitation of fermions to bosonic modes of spin-density and charge-density waves and which prerequisite is the possibility of linearise the single-particle (fermionic) dispersion close to the Fermi level [161].

This linear approximation is pushed to the extreme in the TLL model, meaning that fermions occupy the linearised energy spectrum all the way down to $-\infty$ on the energy scale (also called Dirac sea, Fig. 2.5). In order to apply this assumption, the density of states has to be constant. Broadly speaking, TLL hamiltonian might be written as the sum between a kinetic and an interaction contribution:

$$\mathcal{H}_{\text{TLL}} = \mathcal{H}_0 + \mathcal{H}_{\text{int}}$$

Using the second quantisation formalism, the kinetic part may be written as [153]:

$$\mathcal{H}_0 = \sum_{k;r=R,L} v_F(\epsilon_r k - k_F) c_{k,r}^\dagger c_{k,r}$$

where we have dropped the arrow notation for the wave vector as we are now dealing only with one dimension. Nothing fancy here: $c_{k,r}^\dagger(c_{k,r})$ is the creation (annihilation) operator acting in corresponding Fock space. Because the Fermi surface in 1D consists of just two points at $k = \pm k_F$, we can distinguish states in the right branch ($\epsilon_R = +1$) from those in the left ($\epsilon_L = -1$) branch. The corresponding dispersion relations are:

$$\epsilon_{k,R} = \epsilon_0 + \hbar v_F k$$

$$\epsilon_{k,L} = \epsilon_0 - \hbar v_F k$$

$$\epsilon_0 = \epsilon_F + \hbar v_F k_F$$

Since states in the R branch have positive velocities they are sometimes called *right movers*; conversely, states in the left branch are referred as *left movers*.

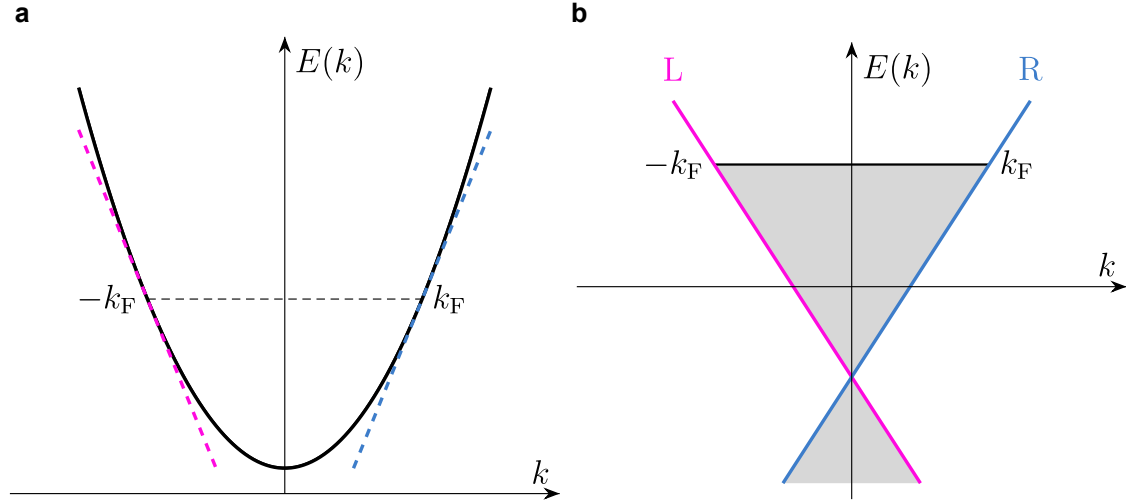


Figure 2.5: The Tomonaga-Luttinger-liquid (TLL) model. **(a)** Free electron parabolic dispersion (black solid curve) and the linearisation carried out according to the TLL model for excitations close the Fermi level (pink and blue dashed lines). **(b)** The linear dispersion used in the TLL model. Pink and blue solid lines correspond, to left-(L) and right (R)-moving electrons, respectively. Shaded area shows the Dirac sea below the Fermi energy. Figure adapted from [162].

The interaction term is where the fun begins. In order to write it down, we shall first introduce a central quantity when one deals with one-dimensional systems. Because the system is subject to quantum fluctuations, even in the zero-temperature limit [163], the physics of the system will be governed by those processes which are close to the Fermi surface. Low-energy processes might be well captured by introducing the density operator*:

$$\rho^\dagger(q) = \sum_k c_{k+q,r}^\dagger c_{k,r} \quad (2.25)$$

The interaction might then be written as:

$$\mathcal{H}_{\text{int}} = \frac{1}{2V} \sum_{k,k',q} V(q) c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k \quad (2.26)$$

*A technicality here. Since the model includes all the negative states, the density operator as it has been written in Eq. 2.25 is infinite as the number of fermions is infinite. One has to introduce the *normal ordering* of operators, meaning that what matters are deviations in the density with respect to the density of the ground state.